

INTELLIGENT REFERRAL DECISION SUPPORT SYSTEM

FHIR, JSON, and the POST Method

A plain-language reference to the healthcare data format used between the Mock Hospital and the IRDSS Interoperability Layer (IOL).

Purpose of this paper

This paper explains what FHIR and JSON mean, which FHIR-style fields are defined by the current IOL server, what data the Mock Hospital sends, and how the HTTP POST method delivers a new referral.

Understanding FHIR and JSON

What is FHIR?

FHIR means **Fast Healthcare Interoperability Resources**. It is a healthcare information standard developed by HL7. It provides agreed names and structures for information that different health systems need to exchange.

Instead of every hospital inventing a different label and layout for the same information, FHIR organizes information into reusable “resources.” Examples include a **Patient** for demographics and an **Encounter** for an interaction involving a patient.

What is JSON?

JSON means **JavaScript Object Notation**. It is a simple text format that computers can create, send, read, and validate. JSON stores information as labeled values. Curly braces contain one object, square brackets contain a list, and a colon separates a label from its value.

FHIR

Defines what the healthcare information means and how it is organized.

JSON

Writes that information in a format that computer systems can exchange.

POST

Delivers the completed JSON to an API to create a new referral.

A small example

```
{
  "resourceType": "Patient",
  "id": "P-000026",
  "gender": "female",
  "birthDate": "1997-05-15"
}
```

Part	Plain-language meaning
resourceType	This tells the receiving system that the object represents a FHIR Patient.
id	This is the hospital's identifier for the patient record.
gender	This contains the recorded gender value.
birthDate	This contains the date of birth using year-month-day format.

Simple comparison: FHIR is like an agreed healthcare form, JSON is the typed version of that form, and POST is the courier that delivers it to the IOL.

FHIR-style structure accepted by the IOL

The IOL server uses Python validation models to check an incoming referral. The main model is named `ReferralRequestSchema`. It currently requires the following top-level fields.

IOL field	Required?	Purpose in the current system
<code>resourceType</code>	Yes	Identifies the referral payload as an Encounter.
<code>status</code>	Yes	Shows the state of the encounter; the Mock Hospital sends <code>planned</code> .
<code>subject</code>	Yes	References the patient using <code>Patient/{id}</code> and may include a display name.
<code>reasonCode</code>	Yes	A list containing one primary referral reason and up to three additional reasons.
<code>patient_record</code>	Yes	Contains the Patient object needed by the current IOL workflow.

Nested definitions used by the IOL

Definition	Fields	Meaning
<code>FHIRSubject</code>	reference, optional display	Connects the referral to the patient.
<code>FHIRReasonCode</code>	text, optional coding	Describes why the patient is being referred.
<code>FHIRCoding</code>	system, code, display	Provides a computer-readable code and a human-readable label.
<code>FHIRPatient</code>	resourceType, id, name, gender, birthDate, extension	Contains the patient demographics and IRDSS-specific information.
<code>FHIRName</code>	family, list of given names	Separates surname from given names.
<code>FHIRExtension</code>	url plus decimal, string, or Boolean value	Adds information needed by IRDSS that is not placed in the basic Patient fields.

Validation applied by the IOL

- The referral must contain between one and four referral reasons.
- Each reason must contain text and must not exceed 255 characters.
- Duplicate referral reasons are rejected.
- The Patient must contain an ID, at least one name entry, a birth date, and an extension list.

Important: `patient_record` is an IRDSS-specific wrapper. It helps the present prototype send an Encounter and Patient together, but it is not a standard field of the official FHIR Encounter resource.

Source in the current IOL: `app/schemas/encounter.py`, `app/schemas/patient_schema.py`, and `app/schemas/fhir_base_schema.py`.

Sample referral payload sent to the IOL

The following sanitized example follows the payload assembled by backend/send_referral.php. It does not contain a real patient or a real security credential.

```
{
  "resourceType": "Encounter",
  "status": "planned",
  "subject": {
    "reference": "Patient/P-000026",
    "display": "Sample Patient"
  },
  "reasonCode": [
    {
      "coding": [{
        "system": "http://irdss.gov.ph/fhir/CodeSystem/referral-reason",
        "code": "NO_AVAILABLE_ICU",
        "display": "No Available Intensive Care Unit"
      }],
      "text": "No Available Intensive Care Unit"
    },
    {
      "coding": [{
        "system": "http://irdss.gov.ph/fhir/CodeSystem/referral-reason",
        "code": "SPECIALIST_UNAVAILABLE",
        "display": "Specialist Unavailable"
      }],
      "text": "Specialist Unavailable"
    }
  ],
  "patient_record": {
    "resourceType": "Patient",
    "id": "P-000026",
    "name": [{
      "family": "Patient",
      "given": ["Sample"]
    }],
    "gender": "female",
    "birthDate": "1997-05-15",
    "extension": [
      { "url": ".../severity", "valueDecimal": 3 },
      { "url": ".../chiefComplaint", "valueString": "Difficulty breathing" },
      { "url": ".../diagnosis", "valueString": "Pneumonia" },
      { "url": ".../vitalHeartRate", "valueDecimal": 90 },
      { "url": ".../vitalOxygenSaturation", "valueDecimal": 95 },
      { "url": ".../hasAllergy", "valueBoolean": true },
      { "url": ".../allergyDetails", "valueString": "Seafood" },
      { "url": ".../consentMethod", "valueString": "ELECTRONIC" }
    ]
  },
  "serviceProvider": {
    "reference": "Organization/FAC-000015",
    "display": "Sample Provincial Hospital"
  }
}
```

Meaning of the data sent by the Mock Hospital

Payload section	Information represented	How IRDSS uses it
Encounter	The planned patient referral event.	Starts the referral workflow.
subject	The patient reference and display name.	Links the referral to the correct patient.
reasonCode	Primary reason followed by optional additional reasons.	Explains the referral and supports analytics.
patient_record	Patient demographics and extensions.	Supplies the clinical and demographic data required for matching and care coordination.
serviceProvider	The referring hospital as an Organization reference.	Sent by the Mock Hospital, but not currently declared in the IOL request validation schema.

Extensions currently produced by the Mock Hospital

Group	Examples
Referral and clinical information	Severity, chief complaint, diagnosis, salary used by the prototype ranking logic, latitude, and longitude.
Contact and demographics	Phone, address, civil status, PhilHealth number, and PhilHealth membership status.
Vital signs	Blood pressure, heart rate, respiratory rate, temperature, oxygen saturation, height, and weight.
Patient indicators	PWD, pregnant, senior citizen, has allergy, and allergy details.
Consent information	Consent status, method, text version, recorded time, witness, signer type, signer name, representative relationship, and document SHA-256 hash.

Why extensions are used

FHIR extensions provide a structured way to add information needed by a particular project. Each extension includes a URL that identifies its meaning and a value such as text, a number, or true/false.

```
{
  "url": "http://irdss.gov.ph/fhir/StructureDefinition/isPregnant",
  "valueBoolean": true
}
```

For use outside the current prototype, the IRDSS extension URLs and their rules should be formally published so another FHIR system can interpret them correctly.

CURRENT

The IOL accepts one combined, FHIR-style referral payload designed for the IRDSS workflow.

FUTURE

Full FHIR exchange should use standalone Patient, Encounter, Condition, Observation, ServiceRequest, Consent, and DocumentReference resources.

How POST sends the referral to the IOL

POST is an HTTP method used when a client asks a server to create or process a new record. In IRDSS, the Mock Hospital uses POST when initiating a referral.



POST request

```

POST /api/v1/referral/initiate HTTP/1.1
Host: <iol-server>:8081
Content-Type: application/json
X-System-Code: SYSTEM-001
X-Key-ID: KEY-2026-001
X-Signature-Timestamp: <current-time>
X-Request-Nonce: <unique-random-value>
X-System-Signature: <rsa-signature>

{ "resourceType": "Encounter", "status": "planned", ... }
  
```

Processing sequence

1. The browser submits ordinary form data to the Mock Hospital PHP backend.
2. The backend reads the selected patient from the local MySQL database.
3. It creates the FHIR-style Encounter and Patient JSON shown in this paper.
4. It calculates a SHA-256 hash of the exact body and signs the request with the hospital's RSA private key.
5. The IOL verifies the sender, timestamp, unique nonce, signature, and request schema.
6. The IOL stores the referral and returns referral identifiers as JSON.

Example successful response

```

{
  "status": "success",
  "message": "Referral initiated successfully",
  "referral_tracking_id": "TRK-2026-000123",
  "referral_id": "ref_35b0cfd39e9f"
}
  
```

Security clarification: The RSA signature verifies the sender and detects changes to the JSON. It does not hide the patient data. A production deployment must also use HTTPS/TLS.

References

- HL7 FHIR R4 JSON Representation — <https://hl7.org/fhir/R4/json.html>
- HL7 FHIR R4 RESTful API — <https://hl7.org/fhir/R4/http.html>
- IOL schemas — `app/schemas/encounter.py`, `patient_schema.py`, and `fhir_base_schema.py`
- Mock Hospital payload and POST client — `backend/send_referral.php` and `backend/config.php`